

Basic I/O Interfacing: 8255 PPI

8255 Programmable Peripheral Interface (PPI) Overview

- Data Bus (D0-D7): These are 8-bit bi-directional buses used to transfer data and are connected to the CPU system bus.
- Handshaking: Pins of Port C are used to provide handshaking signals to transfer data between devices.
- Configuration Code: When A0 and A1 both are set to 1, the 8-bit configuration code is transferred from the CPU to the control register to configure the device.

Port Structure and Flexibility:

- Port A and Port B operate as fixed 8-bit ports. To connect an input or output device to them, we must use their full 8-bit.
- Port C is more flexible. It is split into two independent 4-bit parts: Port C Upper (PC4-PC7) and Port C Lower (PC0-PC3). This allows us to connect separate devices, such as using the upper 4 pins for input and the lower 4 pins for output.

OPERATION MODES

The 8255 operates in two broad modes depending on the D7 bit (MSB):

1. Bit Set/Reset (BSR) Mode: Activated when D7 is 0. Only Port C is utilized in this mode to set or reset individual pins.
2. I/O Mode: Activated when D7 is 1 (Mode definition mode). Ports A, B, and C can all be used in I/O mode.

Port Compatibility by Mode:

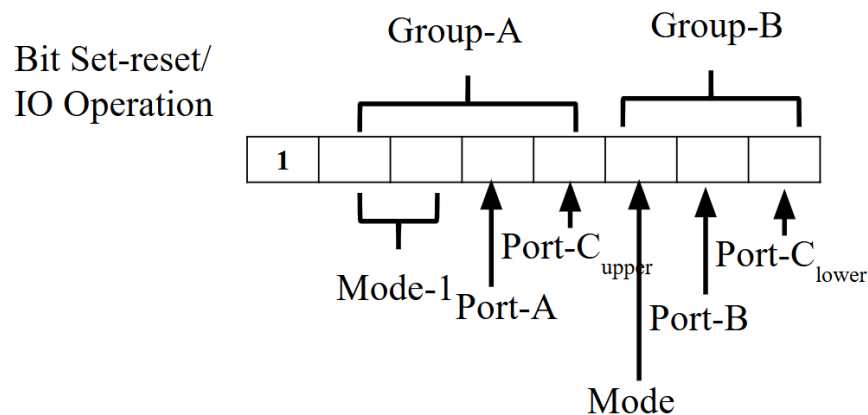
- Port A: Can be programmed in Mode 0, Mode 1, or Mode 2.

- Port B: Can be programmed in Mode 0 or Mode 1 only.
- Port C: Primarily used for simple I/O in Mode 0 or for handshaking in Modes 1 and 2.

I/O MODE DETAILS

- Mode 0 (Simple Input/Output): Ports A, B, and C function independently as simple input or output ports. Used for low level devices. No handshaking is required. Since they operate simply, we could potentially use all 24 pins individually (3 ports * 8 pins = 24 devices).
- Mode 1 (Input/Output with Handshake): Used for uni-directional data transfer where handshaking is required. Ports A and B handle the data transfer, while Port C cannot be in Mode 1 for data itself; instead, it provides the handshaking signals.
- Mode 2 (Bi-directional I/O Data Transfer): This mode allows high-level bi-directional data transfer over a single 8-bit bus using handshaking. Only Port A can be configured in Mode 2, utilizing PC3-PC7 for its handshaking purposes.

I/O PIN MODE (Control Word Format)



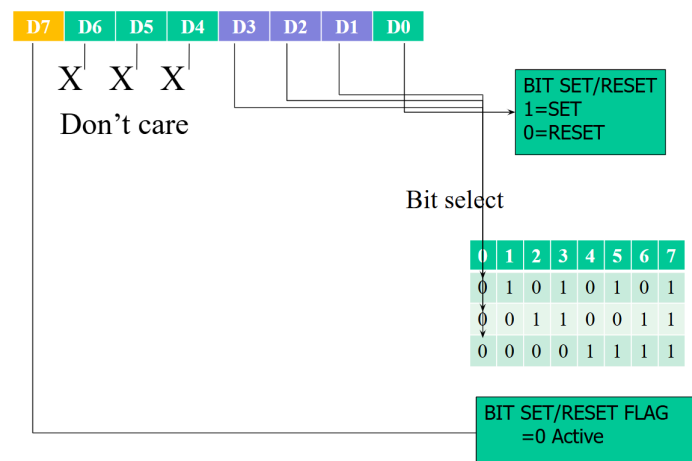
When D7 = 1, the 8-bit control word (D7 to D0) configures the I/O operations:

- D7: Mode selection. must be 1 for I/O mode.
- D6-D5 (Port A Mode):
 - 00 = Mode 0
 - 01 = Mode 1
 - 1x = Mode 2
- D4 (Port A Direction): 1 = Input, 0 = Output.
- D3 (Port C Upper): Controls PC4-PC7 direction. 1 = Input, 0 = Output.
- D2 (Port B Mode): 0 = Mode 0, 1 = Mode 1.
- D1 (Port B Direction): 1 = Input, 0 = Output.
- D0 (Port C Lower): Controls PC3-PC0 direction. 1 = Input, 0 = Output.

BIT SET/RESET (BSR) MODE

- This mode is strictly used for Port C. It allows us to set (1) or reset (0) individual pins of Port C by sending an OUT instruction to the control register. A separate BSR word must be written for each individual bit we wish to change.

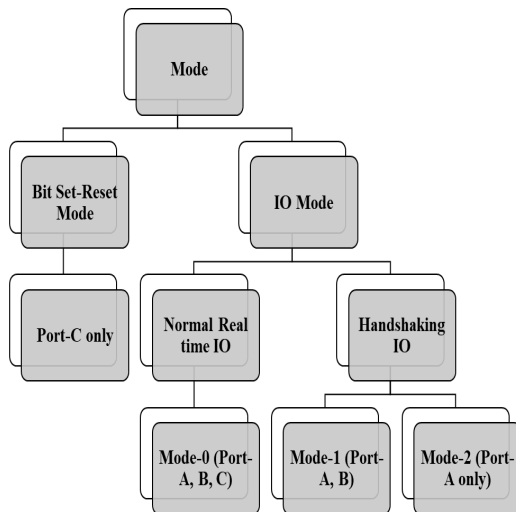
BIT SET/RESET PIN MODE (Control Word Format)



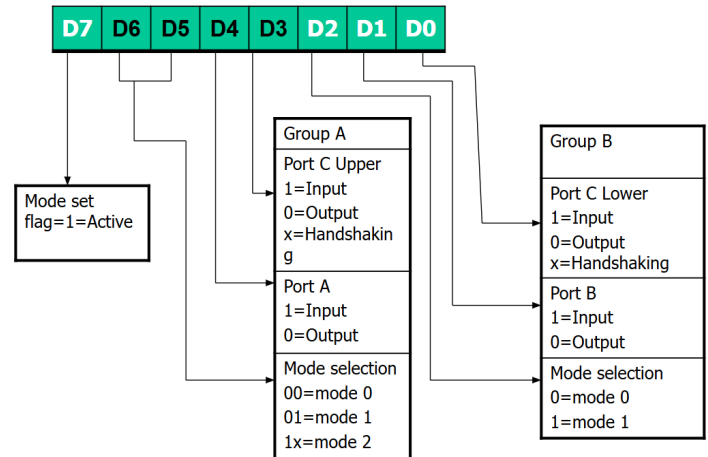
When $D7 = 0$, the 8-bit control word ($D7$ to $D0$) configures the BSR operation:

- $D7$: Mode selection; must be 0 for Bit Set/Reset mode.
- $D6$ - $D4$: Don't care (X).
- $D3$ - $D1$ (Bit Select): These 3 bits act as a binary selector to target one specific pin on Port C ($PC0$ to $PC7$). Example: If $D3$ - $D1$ is 101 (binary for 5), it selects pin $PC5$.
- $D0$ (Set/Reset Flag): Determines the pin's state. 1 = SET, 0 = RESET.

I/O Modes



The mode format for I/O as shown in figure

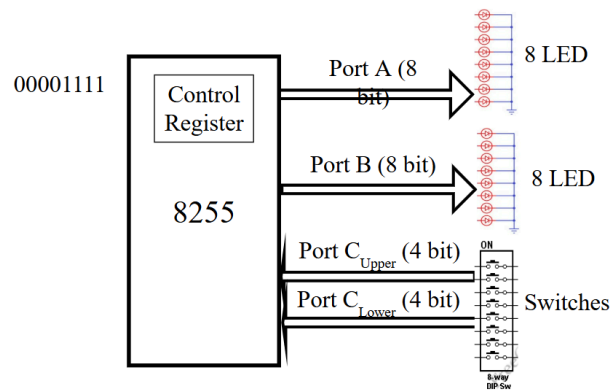


Mode 0: Simple Input/Output

In this mode, Port A, Port B, and Port C are used independently (simple I/O).

Features:

- Outputs are latched
- Inputs are buffered (not latched)
- No handshaking
- No interrupt capability



Given:

D7 → D0 = 1 0 0 0 1 0 0 1

So the bits are:

- D7 = 1
- D6 = 0
- D5 = 0
- D4 = 0
- D3 = 1
- D2 = 0
- D1 = 0
- D0 = 1

That means

- D7 = 1 → I/O Mode
- D6–D5 = 00 → Port A = Mode 0
- D4 = 0 → Port A = Output
- D3 = 1 → Port C Upper = Input

- $D2 = 0 \rightarrow \text{Port B} = \text{Mode 0}$
- $D1 = 0 \rightarrow \text{Port B} = \text{Output}$
- $D0 = 1 \rightarrow \text{Port C Lower} = \text{Input}$

Final Answer:

- Port A $\rightarrow$ Mode 0, Output
- Port B $\rightarrow$ Mode 0, Output
- Port C Upper $\rightarrow$ Input
- Port C Lower $\rightarrow$ Input

If switches are connected in Mode 0 of 82C55, what is the maximum number of switches (input devices) that can be connected?

-In Mode 0, all ports work as simple I/O.

- Port A = 8 pins
- Port B = 8 pins
- Port C = 8 pins

Total = 24 I/O pins.

In Mode 0, the 82C55 provides 24 I/O lines. Therefore, a maximum of 24 input devices (switches) can be connected. If 3 switches are used, then 3 input lines are occupied.

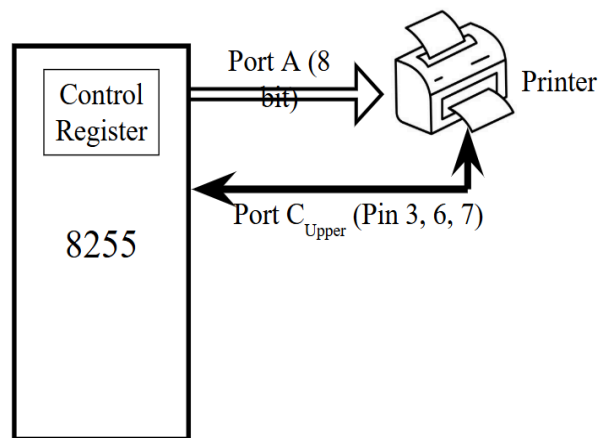
Mode 1: Input/Output with Handshaking

In this mode, data transfer occurs using handshaking signals.

Handshaking is required when connecting high-speed devices with low-speed devices to ensure proper synchronization during data transfer. Because of this, some pins of Port C are used as control (handshaking) signals instead of general I/O.

Concept:

Handshaking is used when two devices have different data transfer speeds.



Example:

- A computer sends data much faster than a printer.
- The printer can only accept data at its own (slower) speed.
- So, the computer sends data only when the printer is ready.
- If the printer is not ready, the computer uses a **handshaking signal** to indicate that valid data is available on the data bus.

Key Points:

- Ensures proper synchronization between devices
- Prevents data loss

Additional Feature:

- Each port uses **three lines from Port C** for handshaking signals

Mode 1 (input) - 82C55 Example

Given:

- Port A → Keyboard (Input)
- Port B → Mouse (Input)
- Mode → Mode 1 (handshaking)

Control Word (D7 → D0):

1 0 1 1 X 1 1 X

Step-by-step Meaning:

- D7 = 1 → I/O Mode
- D6–D5 = 01 → Port A = Mode 1
- D4 = 1 → Port A = Input
- D3 = X → Don't care (used internally for handshake)
- D2 = 1 → Port B = Mode 1
- D1 = 1 → Port B = Input
- D0 = X → Don't care (used internally for handshake)

Handshaking Pins:

Port A (Input Mode 1):

Uses:

- PC4 → $\overline{STB}$ (Strobe)
- PC5 → IBF (Input Buffer Full)
- PC3 → INTR (Interrupt)

Port B (Input Mode 1):

Uses:

- PC2 → $\overline{STB}$ (Strobe)
- PC1 → IBF (Input Buffer Full)
- PC0 → INTR (Interrupt)

Free Pins of Port C:

- Used pins:
 - Port A → PC3, PC4, PC5
 - Port B → PC0, PC1, PC2

Total used = 6 pins

Free pins = PC6, PC7 (2 pins)

Mode 1 Input Timing (82C55) – Explanation

When an input device sends data to the **82C55**, the following sequence happens:

1. $\overline{STB}$ (Strobe) becomes LOW (0)

- The external device places 8-bit data on the bus
- Then it activates $\overline{STB} = 0$
- This tells 82C55: “*Valid data is available*”

2. Data is latched

- The 82C55 accepts and stores the 8-bit data
- Now the input buffer is full

3. $\text{ST}\bar{\text{B}}$ returns HIGH (1)

- After latching, $\text{ST}\bar{\text{B}}$ goes inactive

4. IBF (Input Buffer Full) becomes HIGH

- $\text{IBF} = 1$
- This indicates:
→ “Data has been received, do NOT send more data”

5. INTR (Interrupt) becomes HIGH

- The 82C55 sends an interrupt request to CPU
- Meaning:
→ “Data is ready, CPU can process it”

6. CPU reads the data

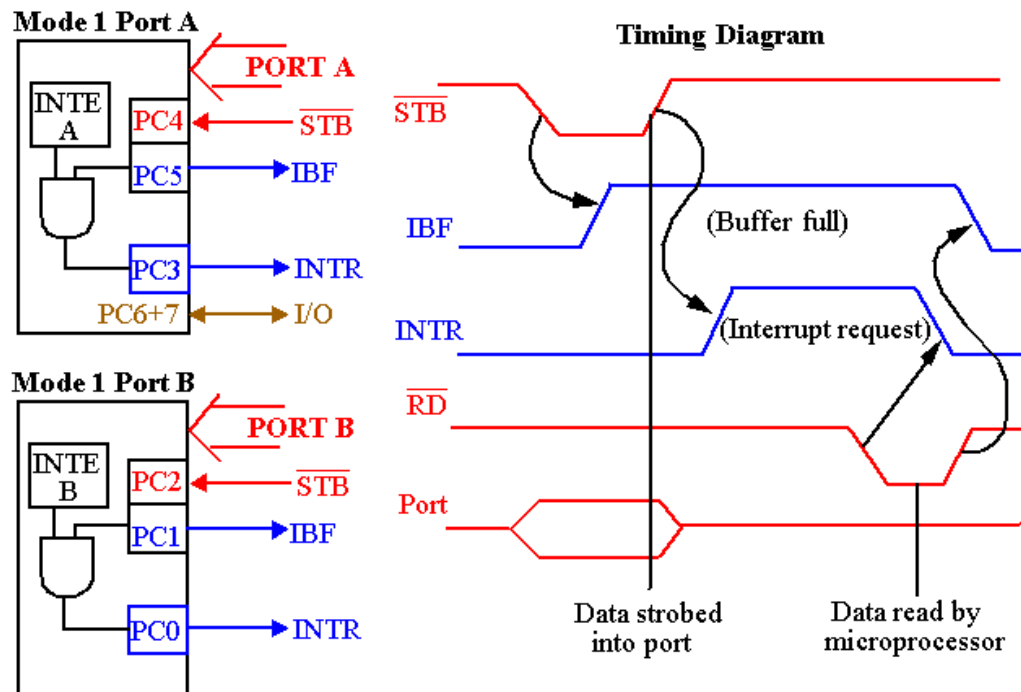
- If CPU is free, it responds
- RD (Read signal) becomes LOW (0)

7. After read operation

- INTR becomes LOW (0)
- IBF resets (data buffer cleared)

Flow Summary:

$\overline{STB} = 0 \rightarrow$ data is latched $\rightarrow \overline{STB} = 1 \rightarrow \text{IBF} = 1 \rightarrow \text{INTR} = 1 \rightarrow$ CPU reads data using $\overline{RD} = 0 \rightarrow$ signals reset



Mode 1 (Output) - 82C55 Example

Given:

- High-level unidirectional output devices
- Port A $\rightarrow$ Display (Output)
- Port B $\rightarrow$ Printer (Output)
- Mode $\rightarrow$ Mode 1 (handshaking output)

Control Word (D7 $\rightarrow$ D0):

1 0 1 0 X 1 0 X

Bit-wise Meaning:

- **D7 = 1** → I/O Mode
- **D6–D5 = 01** → Port A in **Mode 1**
- **D4 = 0** → Port A = **Output**
- **D3 = X** → Don't care (used internally)
- **D2 = 1** → Port B in **Mode 1**
- **D1 = 0** → Port B = **Output**
- **D0 = X** → Don't care (used internally)

Handshaking Signals (Mode 1 Output):

Port A uses:

- PC6 → ACK (Acknowledge)
- PC7 → OBF (Output Buffer Full)
- PC3 → INTR (Interrupt)

Port B uses:

- PC2 → ACK (Acknowledge)
- PC1 → OBF (Output Buffer Full)
- PC0 → INTR (Interrupt)

Free Pins of Port C:

- Used = PC0 - PC2 and PC3, PC6, PC7
- Remaining free pins = **PC4, PC5 (2 pins)**

Mode 1 Output Timing (82C55) - Explanation

When the CPU sends output data to an external device, the following sequence occurs:

1. INTR becomes HIGH

- 82C55 sends an interrupt to CPU
- This indicates:
→ “Output device is ready, send data”

2. CPU activates $\overline{WR} = 0$

- CPU writes data to 82C55
- Data is placed into the output buffer

3. OBF becomes LOW (0)

- Output Buffer Full signal is activated
- This means:
→ “Data is now available for external device”

4. External device checks OBF

- If OBF is active, device knows data is ready
- Then it accepts the data from 82C55

5. ACK signal is sent

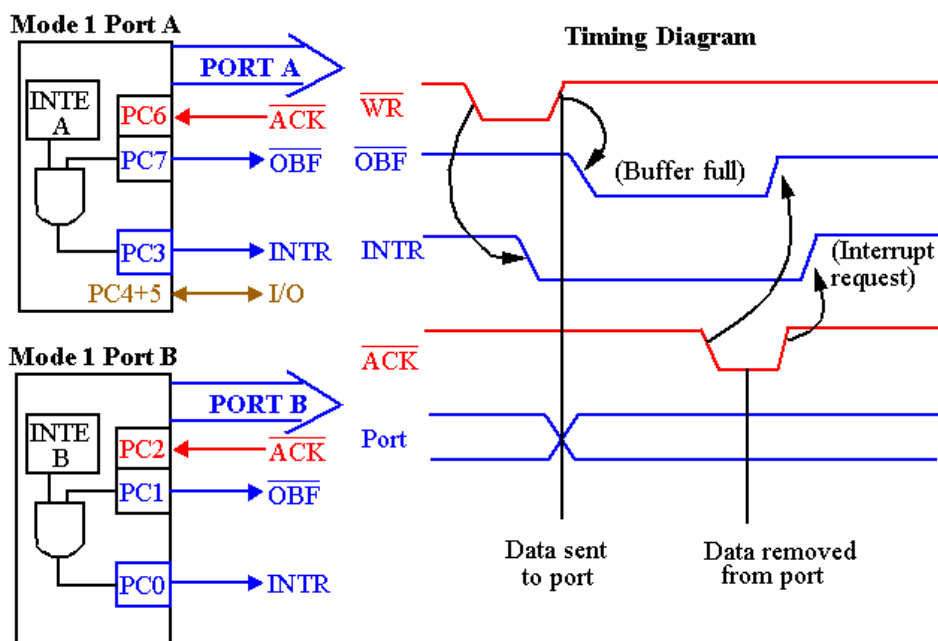
- After receiving data, the external device sends $\text{ACK} = 0$
- This confirms:
→ “Data received successfully”

6. Signals reset

- INTR goes LOW
- $\overline{\text{OBF}}$ returns HIGH
- System becomes ready for next transfer

Flow Summary:

INTR = 1 → CPU sends data using $\overline{\text{WR}} = 0$ → data stored in buffer →
 $\overline{\text{OBF}} = 0$ → device reads data → $\overline{\text{ACK}} = 0$ → signals reset

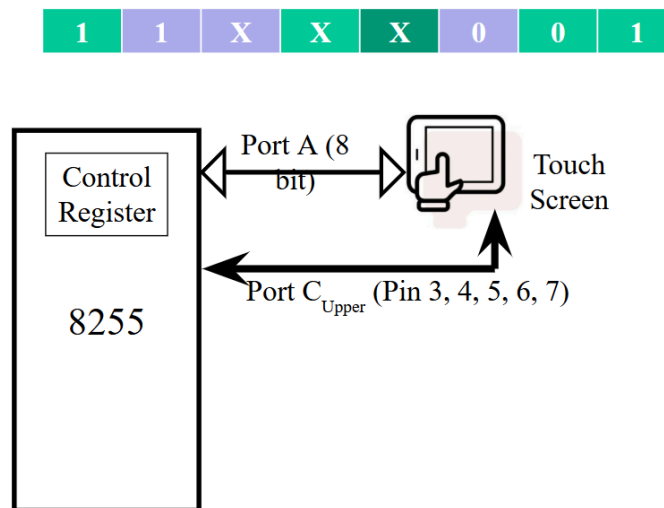


Mode 2: Bi-directional I/O Data Transfer

Mode 2 allows data to be transferred in both directions using a single 8-bit data bus. This transfer is controlled by handshake signals to ensure proper communication.

This mode is available only in **Port A**. In this mode, **Port A** operates as an 8-bit bi-directional input/output port. The control signals required for handshaking are provided by **PC3 to PC7**.

The CPU sends or receives data through Port A whenever needed, with the handshake signals ensuring that the data transfer is properly synchronized.



Configuration 1: 1 Input and 1 Output

- **Port A → Keyboard (Input)**

Handshaking pins used: **PC3, PC4, PC5**

- **Port B → Display (Output)**

Handshaking pins used: **PC3, PC1, PC2**

- **Why is PC3 used in both?**

Because **the operations are different** (input vs output), so the same pin can serve different control purposes.

- **Free pins in Port C:**

3 pins → PC0, PC6, PC7

- **1 1 X 1 X 1 0 X**

Bit Explanation

- $D7 = 1 \rightarrow$ I/O mode enabled
- $D6 D5 = 1X \rightarrow$ Port A Mode 2 (Input with handshaking)
- $D4 = 1 \rightarrow$ Port A = Input (Keyboard)
- $D3 = X \rightarrow$ PC upper (PC4–PC7) used for handshaking (so direction not fixed)
- $D2 = 1 \rightarrow$ Group B Mode 1
- $D1 = 0 \rightarrow$ Port B = Output (Display)
- $D0 = X \rightarrow$ PC lower (PC0–PC3) used for handshaking

Can we connect 2 LEDs and 1 switch in PORT C?

- Yes, we can connect 2 LEDs and 1 switch:
 - **LEDs (Output)** $\rightarrow$ Upper portion $\rightarrow$ **PC6, PC7**
 - **Switch (Input)** $\rightarrow$ Lower portion $\rightarrow$ **PC0**
- **Important Rule:**
 - Lower portion (**PC0–PC3**) $\rightarrow$ should be used for **same type (all input or all output)**
 - Upper portion (**PC4–PC7**) $\rightarrow$ should also be used for **same type (all input or all output)**

1 1 X 1 0 1 0 1

Bit Explanation

- $D7 = 1 \rightarrow$ I/O mode
- $D6 D5 = 1X \rightarrow$ Port A Mode 2
- $D4 = 1 \rightarrow$ Port A = Input (Keyboard)
- $D3 = 0 \rightarrow$ PC upper = Output $\rightarrow$ used for LED (PC6, PC7)

- $D2 = 1 \rightarrow$ Port B Mode 1
- $D1 = 0 \rightarrow$ Port B = Output (Display)
- $D0 = 1 \rightarrow$ PC lower = Input $\rightarrow$ used for Switch (PC0)

Configuration 2: Swapped Devices

- **Port A $\rightarrow$ Display (Output)**
Handshaking pins: **PC3, PC6, PC7**
- **Port B $\rightarrow$ Keyboard (Input)**
Handshaking pins: **PC3, PC1, PC2**
- **Free pins in Port C:**
3 pins $\rightarrow$ PC0, PC4, PC5

Mode 2 (Bidirectional) Configuration

Port A $\rightarrow$ Pendrive (Bidirectional I/O)

- Port A is used for **both input and output**.
- Handshaking requires **5 pins**:
 - **PC3, PC4, PC5, PC6, PC7**
- This is because **bidirectional communication needs more control signals**.

Port B $\rightarrow$ Not used initially

Free Pins in Port C

- Available pins: **PC0, PC1, PC2 $\rightarrow$ Total = 3 pins**

Connecting Devices Using Free Pins

- If we connect **LEDs (Output)**:

- Example control word: **1 1 X X X _ _ 0**
 - Last bit **0** → **Output (LED)**
 - If it were **1** → **Input (Switch)**

Bit Explanation:

- D7 = 1 → I/O mode enabled
- D6–D5 = 1X → Mode 2 (Bidirectional for Port A)
- D4 = X → Don't care (Port A bidirectional in Mode 2)
- D3 = X → Don't care (PC upper used for handshaking)
- D2 = _ → Not used (Port B unused)
- D1 = _ → Not used (Port B unused)
- D0 = 0 → Port C lower = Output → LED

- **We cannot connect LED and Switch together here**
 - Because remaining Port C pins must be used for **same type of operation** (either all input or all output)

Total Free Pins

- Port B (8 pins) + Port C (3 pins) = 11 pins

Possible Connections

- **3 LEDs + 8 Switches possible**
 - Port B → 8 Switches (Input)
 - Port C → 3 LEDs (Output)

Now Using Port B

Port A → Pendrive (Bidirectional)

- Uses: **PC3, PC4, PC5, PC6, PC7**

Port B → Display / Keyboard

- Uses handshaking pins: **PC0, PC1, PC2**

Free Pins in Port C

- No free pins left (0 pins available)

Control Word Example

- **1 1 X X X 1 0 X**

Bit Explanation:

- D7 = 1 → I/O mode
- D6–D5 = 1X → Mode 2
- D4 = X → Don't care (Port A bidirectional)
- D3 = X → PC upper used for handshaking
- D2 = 1 → Port B Mode 1 (handshaking mode)
- D1 = 0 → Port B = Output (Display)
- D0 = X → Don't care (PC lower used for handshaking)